

Exercises 1.7

1. Use a direct proof to show that the sum of two odd integers is even.

Soln: p : x and y are two arbitrary odd integers
 q : $(x+y)$ is even

We shall prove $p \rightarrow q$ is True using a direct proof

Let p be True. $\therefore$ Let x and y are two arbitrary odd integers.

By defn 1, there exists integers k_1 and k_2 such that $x = 2k_1 + 1$
and $y = 2k_2 + 1$.

$$\therefore x + y = (2k_1 + 1) + (2k_2 + 1) = 2(k_1 + k_2 + 1)$$

$\therefore k_1, k_2$ are integers, $k_1 + k_2 + 1$ is an integer.

By defn 1, $x+y$ is an even integer. $\square$

$\therefore q$ is True

2. Use a direct proof to show that the sum of two even integers is even.

Soln: p : x and y are two arbitrary even integers
 q : $(x+y)$ is even.

We shall prove $p \rightarrow q$ is True using a direct proof.

Let p be True. $\therefore$ Let x and y are two arbitrary even integers.

By defn 1, there exists integers k_1 and k_2 such that $x = 2k_1$ and $y = 2k_2$.

$$\therefore x + y = 2k_1 + 2k_2 = 2(k_1 + k_2)$$

$\therefore k_1, k_2$ are integers, $k_1 + k_2$ is an integer.

By defn 1, $x+y$ is an even integer. $\therefore q$ is True $\square$

3. Show that the square of an even number is an even number using a direct proof.